

COMMUNICATION SYSTEMS, SYNCHRONIZATION PROTOCOLS, AND DATA TRANSMISSION ARCHITECTURES

General Communication Architecture

The longitudinal penile physiology monitoring device may incorporate wireless or wired communication systems enabling transmission, synchronization, and exchange of physiologic data with external computational platforms or additional wearable devices. Communication architecture may support real-time monitoring, delayed synchronization, or periodic data transfer depending on operational configuration.

Communication functionality may be integrated directly within the wearable device electronics or distributed across companion devices forming a coordinated monitoring ecosystem.

Wireless Communication Embodiments

Wireless communication technologies may include Bluetooth, Bluetooth Low Energy, ultra-wideband (UWB), near-field communication (NFC), Wi-Fi, cellular communication, body-area network protocols, or future wireless standards capable of transmitting physiologic data securely and efficiently.

Wireless transmission may operate continuously, periodically, or in event-driven modes triggered by detected physiologic transitions, user interaction, or synchronization schedules.

Device Synchronization and Time Alignment

Synchronization protocols may align data streams between multiple sensing devices or computational systems using shared timestamps, clock synchronization methods, beacon signaling, handshake protocols, or post-processing temporal reconstruction techniques. Synchronization may enable multi-node physiologic correlation and coordinated analysis across distributed monitoring devices.

Data Transmission Modes

Data transmission may occur through streaming transmission, batch upload, compressed summary transfer, edge-computed feature transmission, or adaptive hybrid communication strategies balancing bandwidth usage and energy consumption.

Edge processing may allow preliminary analysis to occur locally on the wearable device, reducing communication overhead while preserving essential physiologic insight.

Security and Privacy Protection

Communication systems may incorporate encryption, authentication protocols, secure pairing mechanisms, anonymization procedures, and permission-based access controls to protect sensitive physiologic information during transmission and storage.

Future Communication Technologies

The disclosure expressly includes emerging communication technologies not yet developed at the time of filing, including low-power wide-area networking, implant-to-wearable communication, mesh wearable networks, and other data exchange architectures capable of supporting distributed physiologic monitoring ecosystems.

The communication embodiments described herein are illustrative and non-limiting and encompass present and future transmission systems enabling wearable longitudinal physiologic monitoring.